

Small-field hash-based SNARGs are less sound than conjectured*

Giacomo Fenzi
giacomo.fenzi@epfl.ch
EPFL

Antonio Sanso
antonio.sanso@ethereum.org
Ethereum Foundation

April 9, 2026

Abstract

Hash-based succinct non-interactive arguments (SNARGs) are a widely studied and deployed class of proof systems. The security of practical hash-based SNARGs relies on two combinatorial parameters of its underlying linear code $\mathcal{C}$: a distance-preservation error $\varepsilon_{\text{mca}}(\mathcal{C}, \delta)$ and the list size $|\Lambda(\mathcal{C}, \delta)|$ (both parametrized by a proximity parameter δ).

Optimistically, one might hope that these parameters are bounded all the way to the *capacity regime*: when the proximity parameter δ approaches the minimum distance of the code $\delta(\mathcal{C})$. Perhaps too optimistically, several deployed hash-based SNARGs indeed operate in this regime, and initiatives such as the Ethereum Proximity Prize investigate to which extent soundness is preserved in this setting.

We present a minimal toy protocol whose analysis captures most of the complexity of state-of-the-art hash-based SNARGs, and present a generic attack whose success probability depends on the list size $|\Lambda(\mathcal{C}, \delta)|$.

Further, we investigate the common settings when the code $\mathcal{C}$ is an extension code over a field $\mathbb{F}$ of a base code $\mathcal{C}_{\mathbb{B}}$ over a small base field $\mathbb{B}$. In this setting, we show that classical combinatorial lower bounds on the list-size of the code yields strong attacks that affect the regimes in which hash-based SNARGs operate in practice.

Keywords: hash-based succinct arguments; Reed–Solomon codes; list decoding; proximity gaps conjectures

*A subset of the material in this note also appears in an upcoming survey [ABF26].

1 Introduction

Succinct non-interactive arguments (SNARGs) [Mic00] have emerged as a fundamental cryptographic primitive. SNARGs enable a prover to produce a small certificate that it executed a computation correctly. Surprisingly, this certificate can be exponentially smaller, and more efficient to verify, than the complexity of the original computation. Among SNARGs, those whose security relies purely on the random oracle model [BR93] have been exceedingly successful [AHIV17; BBHR18; BGKS20; BFHVXZ20; DP25; GLSTW23; ACFY24; ZCF24; ACFY25; BCFRRZ25; WHV24; DP24; MZ25; BMNW25a; BMNW25b; BCFW25; BMMS25a; BMMS25b; NA25; GMW25; BFRW25].

These protocols, collectively known as *hash-based SNARGs*, have a number of desirable properties such as no trusted setup, efficient provers that do not need to rely on public key operations, efficient verification and more. This has led to their wide deployment.

Argument size. Compared to other proof systems, hash-based SNARGs suffer from larger argument size, with proofs ranging from few dozens of KiBs to a few MiBs. Hash-based SNARGs rely on linear error correcting codes and in fact modern hash-based SNARGs almost directly inherit the efficiency properties of the error correcting code underlying it (see e.g. [GLSTW23; ACFY25; NA25; BCFW25; BFRW25]). Letting $\lambda \in \mathbb{N}$ be a security parameter and $\mathcal{C}: \mathbb{F}^k \rightarrow \mathbb{F}^n$ be a linear error correcting code, the argument size is dominated by $O(t \cdot \lambda \cdot \log n)$, where $t \in \mathbb{N}$ must be such that

$$(1 - \delta)^t \leq 2^{-\lambda},$$

and $\delta \in (0, 1)$ is the *proximity parameter* the protocol is analyzed under. As $\delta \rightarrow 1$, $t \rightarrow 0$ and thus for minimizing argument size one aims to provide an analysis of a protocol that enables setting δ as large as possible. However, too large of a distance is detrimental to two other quantities that influence the soundness error, and thus a balance must be struck.

Conjectures. When analyzing a hash-based SNARG using a proximity parameter $\delta \in (0, 1)$ two quantities appear in the soundness error:

- *The distance preservation error $\varepsilon_{\text{mca}}(\mathcal{C}, \delta)$.* This term captures the probability that when taking a random linear combination of words f_1, f_2 that are not both δ -close to the code $\mathcal{C}$ the combination ends up δ -close to $\mathcal{C}$. This family of errors captures a number of notions such as agreement [RVW13], correlated agreement [BCIKS20] and mutual correlated agreement [ACFY25]. For soundness, we require that $\varepsilon_{\text{mca}} \leq 2^{-\lambda}$.
- *The list size $|\Lambda(\mathcal{C}, \delta)|$.* This term bounds the maximal number of codewords that can be δ -close to a word f . Typically, the soundness of a protocol has a term of the form $|\Lambda(\mathcal{C}, \delta)| / |\mathbb{F}|$, which is then also required to be smaller than $2^{-\lambda}$ for soundness.

Optimistically, one could hope that for proximity parameter δ as large as the minimum distance of the code $\delta(\mathcal{C})$ while keeping both these quantities under control. However, the current analyses (specialized to the Reed–Solomon code [RS60]) only yield provable guarantees to much smaller distances such as when $\delta \approx 1 - \sqrt{1 - \delta(\mathcal{C})}$ (the Johnson bound). Establishing whether Reed–Solomon codes can provide soundness for distances between the Johnson bound and the “capacity bound” $\delta(\mathcal{C})$ is the objective of a recent initiative [E25] by the Ethereum Foundation. See [ABF26] for a survey on what is known.

1.1 Our results

In this note, we establish information-theoretic limitations on the soundness of hash-based protocols commonly deployed in practice. Our contributions can be summarized as follows:

- We formalize a simple “toy” IOPP that abstracts core components of several hash-based SNARGs, and recall (from companion work [ABF26]) that its soundness analysis depends only on a distance-preservation term $\varepsilon_{\text{mca}}(\mathcal{C}, \delta)$, the list size $|\Lambda(\mathcal{C}, \delta)|$ at radius δ (Lemma 4.3) and a proximity term $(1 - \delta)^t$.
- We show that this dependence is essentially tight by giving a list-decoding based attack whose success probability is lower bounded in terms of $|\Lambda(\mathcal{C}, \delta)|$ and $(1 - \delta)^t$ (Lemma 4.4). In particular, large lists at radius δ necessarily translate into non-negligible soundness error.
- We then specialize to the practically relevant setting of extension codes, where $\mathcal{C}$ over $\mathbb{F}$ is obtained from a base code $\mathcal{C}_{\mathbb{B}}$ over a smaller field $\mathbb{B}$. We observe that the list size of $\mathcal{C}$ is lower bounded by that of $\mathcal{C}_{\mathbb{B}}$ (Section 4 and Section 3), so combinatorial lower bounds over the small alphabet $\mathbb{B}$ imply soundness limitations of the proof system over $\mathbb{F}$.
- Finally, we instantiate our attack for concrete Reed–Solomon parameters that resemble deployed SNARG systems (Section 4.3), and show that the success probability of our attack is $\approx 2^{-116.5}$ for typical choices of parameters which were previously conjectured to provide 128 bits of security (e.g. rate $\rho = 1/2$, $t = 128$, and a 31-bit base field).

1.2 Related work

An exciting recent line of work [DG25; CS25] has established that the “capacity regime” for δ might indeed be too optimistic for Reed–Solomon codes. Let $\mathcal{C}$ be a Reed–Solomon code over a field $\mathbb{F}$ and of rate ρ (and thus minimal distance $\delta(\mathcal{C}) = 1 - \rho$).

- [DG25] study proximity gaps for Reed–Solomon codes and falsify the conjecture in [BCIKS20] that those codes exhibit proximity cap up to capacity. More formally, they prove that for every fixed constant $c > 0$ and all large enough block lengths n , there exist infinitely many parameter choices with

$$\delta \gtrsim \delta(\mathcal{C}) - \frac{\rho \cdot \log(e/\rho)}{\log |\mathbb{F}|}$$

such that the proximity error $\varepsilon_{\text{mca}}(\mathcal{C}, \delta)$ is at least $n^{-c}/|\mathbb{F}|$.

- [CS25, Sec. 3.1] shows that the “up-to-capacity” list-decodability conjecture is false (for general linear codes): using Elias’ bound [Eli57], they prove that whenever the rate ρ and distance parameter δ satisfy

$$H_{|\mathbb{F}|}(\delta) > \delta(\mathcal{C})$$

(where $H_{|\mathbb{F}|}$ is the $|\mathbb{F}|$ -ary entropy), any (δ, L) -list-decodable code must have *exponential* list size $L \geq |\mathbb{F}|^{\Omega(\eta n)}$ for some constant $\eta > 0$. Further, for Reed–Solomon code they rule out any “up-to-capacity” proximity-gap or mutual correlated-agreement conjecture asserting $\varepsilon_{\text{mca}}(\mathcal{C}, \delta) \ll 1$ in the regime $\delta \approx 1 - \rho$, and they propose modified conjectures where the capacity condition $\delta \leq 1 - \rho - \eta$ is replaced by the list-decoding capacity constraint $H_{|\mathbb{F}|}(\delta) \leq 1 - \rho - \eta$.

2 Preliminaries

We use the following convention and notation: (i) The “hat” symbol over a function (e.g. $\hat{p}$) denotes a polynomial. (ii) We interchangeably interpret vectors $\mathbf{f} \in \Sigma^n$ as functions $f: [n] \rightarrow \Sigma$. (iii) The maxima of a tuple is interpreted to be the maxima of the individual components i.e. $\max_{\delta \in S}(a(\delta), b(\delta)) = (\max_{\delta \in S} a(\delta), \max_{\delta \in S} b(\delta))$.

We recall the *polynomial identity lemma*.

Lemma 2.1 (Ore–DeMillo–Lipton–Schwartz–Zippel lemma). *Let $\hat{p} \in \mathbb{F}^{<d}[X_1, \dots, X_m]$ be a non-zero polynomial. Then,*

$$\Pr_{\mathbf{v} \leftarrow \mathbb{F}^m} [\hat{p}(\mathbf{v}) = 0] \leq \frac{m \cdot (d-1)}{|\mathbb{F}|}.$$

We also recall zero-evaders, which are functions that expand a short seed into a long vector, guaranteeing that the inner product with any non-zero vector will be non-zero with high probability.

Definition 2.2. A **zero-evader** over $\mathbb{F}$ with error $\varepsilon_{\text{zero}}$ is a function $\text{ZE}: D_{\text{ZE}} \rightarrow \mathbb{F}^m$ such that

$$\forall \mathbf{v} \in \mathbb{F}^m \setminus \{\mathbf{0}\}, \Pr_{\rho \leftarrow D_{\text{ZE}}} [\langle \text{ZE}(\rho), \mathbf{v} \rangle = 0] \leq \varepsilon_{\text{zero}}.$$

Lemma 2.1 implies the existence of a large class of polynomial zero-evaders, the most relevant of which for this note is the identity zero-evader (which samples a vector $\mathbf{u} \in \mathbb{F}^n$ and has error $\varepsilon_{\text{zero}} = \frac{1}{|\mathbb{F}|}$). We show that, for any set, there must exist some zero-evader which maps elements of the set to a set of size that is not too small.

Claim 2.3. Let $\text{ZE}: D_{\text{ZE}} \rightarrow \mathbb{F}^k$ be a zero-evader. For any $S \subseteq \mathbb{F}^k$ there exists some $\rho \in D_{\text{ZE}}$ such that

$$|\{\langle \mathbf{f}, \text{ZE}(\rho) \rangle : \mathbf{f} \in S\}| \geq \frac{|S|}{(|S| - 1) \cdot \varepsilon_{\text{zero}} + 1}.$$

Proof. For $\rho \in D_{\text{ZE}}$, we let $S_\rho = \{\langle \mathbf{f}, \text{ZE}(\rho) \rangle : \mathbf{f} \in S\}$ denote the set of images of S under the linear map $\langle \cdot, \text{ZE}(\rho) \rangle$. Fix $\rho \in D_{\text{ZE}}$, and for $\mu \in S_\rho$ let $U_\mu^\rho = \{\mathbf{f} \in S : \langle \mathbf{f}, \text{ZE}(\rho) \rangle = \mu\}$ denote the fibers of elements of S_ρ . Further let $C_\rho = \left\{(\mathbf{f}, \mathbf{g}) \in \binom{S}{2} : \langle \mathbf{f} - \mathbf{g}, \text{ZE}(\rho) \rangle = 0\right\}$ be the set of (distinct) colliding elements of S under ρ . Note in particular that $\mathbb{E}_\rho[|C_\rho|] \leq \binom{|S|}{2} \cdot \varepsilon_{\text{zero}}$. Clearly, $|S| = \sum_{\mu \in S_\rho} |U_\mu^\rho|$, and further (since each $\mu \in S_\rho$ contributes $\binom{|U_\mu^\rho|}{2}$ colliding pairs) it must be that

$$|C_\rho| = \sum_{\mu \in S_\rho} \binom{|U_\mu^\rho|}{2} = \frac{1}{2} \sum_{\mu \in S_\rho} (|U_\mu^\rho| - 1) \cdot |U_\mu^\rho| = \frac{1}{2} \left(-|S| + \sum_{\mu \in S_\rho} |U_\mu^\rho|^2 \right).$$

Rearranging, multiplying through by $|S_\rho|$ (and applying Cauchy–Schwarz), we obtain that

$$|S_\rho| \cdot (2|C_\rho| + |S|) = |S_\rho| \cdot \sum_{\mu \in S_\rho} |U_\mu^\rho|^2 \geq \left(\sum_{\mu \in S_\rho} |U_\mu^\rho| \right)^2 = |S|^2,$$

and thus $|S_\rho| \geq \frac{|S|^2}{2|C_\rho| + |S|}$. By taking expectations, and by Jensen’s we get that

$$\mathbb{E}_\rho[|S_\rho|] \geq \mathbb{E}_\rho \left[\frac{|S|^2}{2|C_\rho| + |S|} \right] \geq \frac{|S|^2}{2 \cdot \mathbb{E}_\rho[|C_\rho|] + |S|} \geq \frac{|S|^2}{2 \cdot \binom{|S|}{2} \cdot \varepsilon_{\text{zero}} + |S|}.$$

Rearranging, we get the final bound, and this guarantees the existence of a ρ with the desired properties. $\square$

2.1 Information theory

We recall the definition of the entropy function with respect to a set.

Definition 2.4. For $q \in \mathbb{N}$, the q -ary entropy function is the function $H_q: [0, 1] \rightarrow \mathbb{R}$ defined as

$$H_q(x) = x \cdot \log_q(q-1) - x \log_q(x) - (1-x) \cdot \log_q(1-x).$$

For $x \in (0, 1 - \frac{1}{q})$ we have that $x < H_q(x) \leq x - \frac{1}{\log q}$ (see e.g. [CS25] for a proof).

The (fractional) Hamming distance between two vectors is the fraction of points in which they differ.

Definition 2.5. Let $f, g: S \rightarrow \Sigma$. The (fractional) Hamming distance between f and g is

$$\Delta_T(f, g) = \Pr_{i \leftarrow S}[f(i) \neq g(i)].$$

For a set $\mathcal{C}$, we write $\Delta(f, \mathcal{C}) = \min_{g \in \mathcal{C}} \Delta(f, g)$.

The volume of the Hamming ball of distance δ around a point in space is the number of points in the space whose Hamming distance to the point is at most δ :

Definition 2.6. For any $q, n \in \mathbb{N}$ and $\delta \in (0, 1)$, we denote the **volume of the Hamming ball** of radius δ over the space q^n is

$$\text{Vol}_q(\delta, n) = \sum_{i=0}^{\lfloor \delta \cdot n \rfloor} \binom{n}{i} \cdot (q-1)^i.$$

2.2 Error correcting codes

We recall the definition of (linear) error correcting codes.

Definition 2.7. An error correcting code $\mathcal{C}$ over an alphabet Σ with block length n is a subset $\mathcal{C} \subseteq \Sigma^n$. The **minimum distance** of $\mathcal{C}$ is $\delta(\mathcal{C}) := \min_{f, g \in \mathcal{C}} \Delta(f, g)$, and the **rate** of $\mathcal{C}$ is $\rho(\mathcal{C}) := \log_{|\Sigma|} |\mathcal{C}|/n$. The code $\mathcal{C}$ is **linear** if $\Sigma = \mathbb{F}$ where $\mathbb{F}$ is a finite field, and $\mathcal{C}$ is a linear subspace of $\mathbb{F}^n$.

We also write $\mathcal{C}: \Sigma^k \rightarrow \Sigma^n$ by fixing an embedding from $\mathcal{C}$ (where $k := \log_{|\Sigma|} |\mathcal{C}|$). It is easy to see by definition that, for a code $\mathcal{C} \subseteq \Sigma^n$, for every word $f \in \Sigma^n$ there is at most one codeword $u \in \mathcal{C}$ with $\Delta(f, u) < \delta(\mathcal{C})/2$. For this reason, we call $\delta(\mathcal{C})/2$ the “unique-decoding radius”. We now extend this definition to consider the number of codewords within arbitrary radius δ :

Definition 2.8. Let $\mathcal{C}$ be an error correcting code over alphabet Σ with block length n . For any $f: [n] \rightarrow \Sigma$ and $\delta \in (0, 1)$, we define the **list around f at distance δ** as

$$\Lambda(\mathcal{C}, f, \delta) := \{g \in \mathcal{C} \mid \Delta(f, g) \leq \delta\}.$$

Further, we write $|\Lambda(\mathcal{C}, \delta)| := \max_f |\Lambda(\mathcal{C}, f, \delta)|$.

An interleaved code is constructed by bundling together multiple codewords of another code:

Definition 2.9. Let $\mathcal{C}$ be an error correcting code over alphabet Σ with block length n , and let $m \in \mathbb{N}$. The m -**interleaved code** $\mathcal{C}^{\equiv m}$ is the code over alphabet Σ^m with block length n defined as

$$\mathcal{C}^{\equiv m} := \{(u_1, \dots, u_m) \mid \forall i \in [m] : u_i \in \mathcal{C}\} \subseteq (\Sigma^m)^n.$$

That is, a codeword $u \in \mathcal{C}^{\equiv m}$ consists of $m \cdot n$ symbols of Σ , arranged as a $m \times n$ size matrix where each row corresponds to a codeword of $\mathcal{C}$ (and interpreted as a vector of length n over the alphabet Σ^m).

Clearly, for any $m \in \mathbb{N}$ and $\delta \in (0, 1)$ it holds that $|\Lambda(\mathcal{C}, \delta)| \leq |\Lambda(\mathcal{C}^{\equiv m}, \delta)| \leq |\Lambda(\mathcal{C}, \delta)|^m$. In fact the list-size of an interleaved code at some distance is larger than the list-size of the base code by a factor *independent of the interleaving factor* (but which depends on the distance δ considered).

Lemma 2.10 ([GGR11]). Let $\mathcal{C}$ be a code, let $\delta \in [0, \delta(\mathcal{C})]$ and $\eta := \delta(\mathcal{C}) - \delta$. Let $b := \left\lceil \frac{\delta}{\eta} \right\rceil$ and $r := \left\lceil \log \frac{\delta(\mathcal{C})}{\eta} \right\rceil$. For every $m \geq 1$, $|\Lambda(\mathcal{C}^m, \delta)| \leq \binom{b+r}{r} \cdot |\Lambda(\mathcal{C}, \delta)|^r$.

2.3 Reed–Solomon codes

The main body of this note will center around a specific error-correcting code known as the Reed–Solomon code.

Definition 2.11. Let $\mathbb{F}$ be a finite field, $\mathcal{L} \subseteq \mathbb{F}$ and $k \in \mathbb{N}$. The **Reed–Solomon code** over alphabet $\mathbb{F}$ with message length k and evaluation domain $\mathcal{L}$ is defined as

$$\text{RS}[\mathbb{F}, \mathcal{L}, k] := \left\{ f : \mathcal{L} \rightarrow \mathbb{F} \mid \exists \hat{f} \in \mathbb{F}^{<k}[X], \forall x \in \mathcal{L} : \hat{f}(x) = f(x) \right\}.$$

We let $n := |\mathcal{L}|$ denote the block length of the code and $\rho := \frac{k}{n}$ the rate. The Reed–Solomon code has minimum distance $\delta(\text{RS}[\mathbb{F}, \mathcal{L}, k]) = 1 - \frac{k}{n} = 1 - \rho$.

Note that we interpret codewords in $\text{RS}[\mathbb{F}, \mathcal{L}, k]$ as functions $f : \mathcal{L} \rightarrow \mathbb{F}$ by making use of the bijection from $[n]$ to $\mathcal{L}$.

The most common scenario to use Reed–Solomon codes in applications is over a domain $\mathcal{L}$ which is highly structured, called a “smooth” domain.

Definition 2.12. We say that a set $\mathcal{L} \subseteq \mathbb{F}$ is **smooth** if it is a multiplicative coset of $\mathbb{F}^*$ whose order is a power of two.

2.4 Extension fields

We review definitions about extension fields and codes defined over extension fields.

Definition 2.13. A tuple $(\mathbb{B}, \mathbb{F}, e, \psi, \varphi)$ is a **extension field presentation** if:

- $\mathbb{B}, \mathbb{F}$ are fields;
- $\psi : \mathbb{B} \hookrightarrow \mathbb{F}$ is an injective field homomorphism;
- $e \in \mathbb{N}$ is the dimension of $\mathbb{F}$ as a $\mathbb{B}$ -vector space;
- $\varphi : \mathbb{F} \rightarrow \mathbb{B}^e$ is a $\mathbb{B}$ -linear isomorphism.

For each $i \in [e]$, $\varphi_i : \mathbb{F} \rightarrow \mathbb{B}$ denotes the i -th component of φ . An extension field presentation $(\mathbb{B}, \mathbb{F}, e, \psi, \varphi)$ is **systematic** if, for every $x \in \mathbb{B}$, $\varphi(\psi(x)) = (x, 0, \dots, 0)$.

We also apply ψ to vectors of base field elements and $\varphi, \varphi_1, \dots, \varphi_e$ to vectors of extension field elements by applying each of these maps componentwise.

An extension code defines a code over a field extension for a linear code over the base field.

Definition 2.14. *Let $\mathcal{C}_{\mathbb{B}}: \mathbb{B}^k \rightarrow \mathbb{B}^n$ be a linear code. The **extension code of $\mathcal{C}_{\mathbb{B}}$ over $\mathbb{F}$** (with respect to the extension field presentation $(\mathbb{B}, \mathbb{F}, e, \psi, \varphi)$) is the linear code $\mathcal{C}_{\mathbb{F}}: \mathbb{F}^k \rightarrow \mathbb{F}^n$ defined as*

$$\mathcal{C}_{\mathbb{F}}(\mathbf{v}) := \varphi^{-1}\left(\mathcal{C}_{\mathbb{B}}(\varphi_1(\mathbf{v})), \dots, \mathcal{C}_{\mathbb{B}}(\varphi_e(\mathbf{v}))\right).$$

If the extension field presentation is systematic, then $\mathcal{C}_{\mathbb{F}}(\psi(\mathbf{v})) = \psi(\mathcal{C}_{\mathbb{B}}(\mathbf{v}))$ for any $\mathbf{v} \in \mathbb{B}^k$.

The distance of the extension code $\mathcal{C}_{\mathbb{F}}$ equals the distance of the given code $\mathcal{C}_{\mathbb{B}}$, i.e., $\delta(\mathcal{C}_{\mathbb{B}}) = \delta(\mathcal{C}_{\mathbb{F}})$ [DP25, Thm 3.2].

3 List decoding

We recall a classical lower bound on the (combinatorial) list-decoding of any code, due to [Eli57].

Lemma 3.1. *Let $\mathcal{C}: \Sigma^k \rightarrow \Sigma^n$ be a code and let $q := |\Sigma|$. For any $\delta \in (0, 1)$ it holds that*

$$\mathbb{E}_{\mathbf{v} \leftarrow \Sigma^n} [|\Lambda(\mathcal{C}, \mathbf{v}, \delta)|] = \frac{\text{Vol}_q(\delta, n)}{q^{n-k}}.$$

Proof.

$$\begin{aligned} \mathbb{E}_{\mathbf{v} \leftarrow \Sigma^n} [|\Lambda(\mathcal{C}, \mathbf{v}, \delta)|] &= \sum_{u \in \mathcal{C}} \Pr_{\mathbf{v} \leftarrow \Sigma^n} [\Delta(\mathbf{v}, u) \leq \delta] \\ &= \sum_{u \in \mathcal{C}} \frac{\text{Vol}_q(\delta, n)}{q^n} \\ &= \frac{\text{Vol}_q(\delta, n)}{q^{n-k}}. \end{aligned}$$

□

By the probabilistic method applied to Lemma 3.1, there must exists a $\mathbf{v}$ attaining at least that list-size, and so we obtain the following corollary.

Corollary 3.2. *Let $\mathcal{C}: \Sigma^k \rightarrow \Sigma^n$ be a code and let $q := |\Sigma|$. For any $\delta \in (0, 1)$ it holds that*

$$|\Lambda(\mathcal{C}, \delta)| \geq \frac{\text{Vol}_q(\delta, n)}{q^{n-k}}.$$

Using an estimate on the size of $\text{Vol}_q(\delta, n)$, one can lower bound the size of the list, we do so by using the estimate of [MS77].

Corollary 3.3. *Let $\mathcal{C}: \Sigma^k \rightarrow \Sigma^n$ be a code of rate ρ and let $q := |\Sigma|$. For any $\delta \in (0, 1)$,*

$$|\Lambda(\mathcal{C}, \delta)| \geq \frac{q^{n\eta}}{\sqrt{8 \cdot n \cdot \delta \cdot (1 - \delta)}},$$

where $\eta := \rho - 1 + H_q(\delta)$.

Proof. By [MS77] it holds that $\text{Vol}_q(\delta, n) \geq \frac{q^{n \cdot H_q(\delta)}}{\sqrt{8 \cdot n \cdot \delta \cdot (1 - \delta)}}$. Then,

$$|\Lambda(\mathcal{C}, \delta)| \geq \frac{\text{Vol}_q(\delta, n)}{q^{n(1-\rho)}} \geq \frac{q^{H_q(\delta) \cdot n}}{\sqrt{8 \cdot n \cdot \delta \cdot (1 - \delta)} \cdot q^{n(1-\rho)}} = \frac{q^{n \cdot (\rho - (1 - H_q(\delta)))}}{\sqrt{8 \cdot n \cdot \delta \cdot (1 - \delta)}}.$$

□

Remark 3.4. More accurate estimates on $\text{Vol}_q(\delta, n)$ (such as those in [DG25]) will yield improved (provable) success probability for the attacks in Corollary 4.5. For the purposes of this note, the rougher bounds suffices.

The list size of the extension code is the list size of the e -interleaved base code (which in turn is related to the list size of the base code by Lemma 2.10). The result is due to [BCFW25, Lemma D.3] and we recall the proof for completeness.

Lemma 3.5. *Let $\mathcal{C}_{\mathbb{B}}: \mathbb{B}^k \rightarrow \mathbb{B}^n$ be a linear code and $(\mathbb{B}, \mathbb{F}, e, \psi, \varphi)$ be an extension field presentation. For every $\delta \in (0, 1)$, $|\Lambda(\mathcal{C}_{\mathbb{F}}, \delta)| = |\Lambda(\mathcal{C}_{\mathbb{B}}^e, \delta)|$.*

Proof. We prove two inequalities.

- $|\Lambda(\mathcal{C}_{\mathbb{F}}, \delta)| \leq |\Lambda(\mathcal{C}_{\mathbb{B}}^e, \delta)|$.

Let $f: \mathcal{L} \rightarrow \mathbb{F}$ and $u \in \Lambda(\mathcal{C}_{\mathbb{F}}, f, \delta)$. For every $i \in [e]$, define $f_i := \varphi_i \circ f$ and $u_i := \varphi_i \circ u$. We show that $(u_i)_{i \in [e]} \in \Lambda(\mathcal{C}_{\mathbb{B}}^e, (f_i)_{i \in [e]}, \delta)$. For every $i \in [e]$, observe that $u_i \in \mathcal{C}_{\mathbb{B}}$. Let $\mathbf{v} \in \mathbb{F}^k$ be such that $u = \mathcal{C}_{\mathbb{F}}(\mathbf{v})$. Then,

$$u_i = \varphi_i \circ u = \varphi_i \circ \mathcal{C}_{\mathbb{F}}(\mathbf{v}) = \varphi_i \circ \varphi^{-1}(\mathcal{C}_{\mathbb{B}}(\varphi_1(\mathbf{v})), \dots, \mathcal{C}_{\mathbb{B}}(\varphi_e(\mathbf{v}))) = \mathcal{C}_{\mathbb{B}}(\varphi_i(\mathbf{v})).$$

Since $\Delta(f, u) \leq \delta$, there exists $S \subseteq \mathcal{L}$ with $|S| \geq (1 - \delta) \cdot n$ such that $f(x) = u(x)$ for every $x \in S$. Then, for every $x \in S$ and $i \in [e]$, $f_i(x) = \varphi_i \circ f(x) = \varphi_i \circ u(x) = u_i(x)$. We conclude that $\Delta((f_i)_{i \in [e]}, (u_i)_{i \in [e]}) \leq \delta$.

- $|\Lambda(\mathcal{C}_{\mathbb{F}}, \delta)| \geq |\Lambda(\mathcal{C}_{\mathbb{B}}^e, \delta)|$.

Let $(f_i)_{i \in [e]}: \mathcal{L} \rightarrow \mathbb{B}^e$ and $(u_i) \in \Lambda(\mathcal{C}_{\mathbb{B}}^e, (f_i)_{i \in [e]}, \delta)$. Define $f := \varphi^{-1} \circ (f_1, \dots, f_e)$ and $u := \varphi^{-1} \circ (u_1, \dots, u_e)$. We show that $u \in \Lambda(\mathcal{C}_{\mathbb{F}}, f, \delta)$. Observe that $u \in \mathcal{C}_{\mathbb{F}}$: for every $i \in [e]$, let $\mathbf{v}_i$ be such that $u_i = \mathcal{C}_{\mathbb{B}}(\mathbf{v}_i)$ and define $\mathbf{v} := \varphi^{-1}(\mathbf{v}_1, \dots, \mathbf{v}_e)$; then,

$$\mathcal{C}_{\mathbb{F}}(\mathbf{v}) = \varphi^{-1}(\mathcal{C}_{\mathbb{B}}(\varphi_1(\mathbf{v})), \dots, \mathcal{C}_{\mathbb{B}}(\varphi_e(\mathbf{v}))) = \varphi^{-1}(\mathcal{C}_{\mathbb{B}}(\mathbf{v}_1), \dots, \mathcal{C}_{\mathbb{B}}(\mathbf{v}_e)) = \varphi^{-1}(u_1, \dots, u_e) = u.$$

Since $\Delta((f_i)_{i \in [e]}, (u_i)_{i \in [e]}) \leq \delta$, there exists $S \subseteq \mathcal{L}$ with $|S| \geq (1 - \delta) \cdot n$ such that $f_i(x) = u_i(x)$ for every $x \in S$ and $i \in [e]$. Since $f(x) = \varphi^{-1}(f_1(x), \dots, f_e(x)) = \varphi^{-1}(u_1(x), \dots, u_e(x)) = u(x)$ for every $x \in S$, we conclude that $\Delta(f, u) \leq \delta$.

□

As a direct consequence of Lemma 3.5, we obtain the following corollary.

Corollary 3.6. *Let $\mathcal{C}: \mathbb{F}^k \rightarrow \mathbb{F}^n$ be the extension code of a linear code $\mathcal{C}_{\mathbb{B}}: \mathbb{B}^k \rightarrow \mathbb{B}^n$. Then for any $\delta \in (0, 1)$ we have that $|\Lambda(\mathcal{C}, \delta)| \geq |\Lambda(\mathcal{C}_{\mathbb{B}}, \delta)|$.*

4 Toy problem

In this section, we describe a simple protocol that captures most of the complexities of the soundness analysis of more complex protocols such as [ACFY25; NA25; BCFW25].

4.1 Protocol

The protocol considers a relation consisting of codewords whose underlying messages satisfy a given linear constraint. We refer to this relation to be about *constrained codes*.

Definition 4.1. *We define the toy problem relation.*

$$\mathcal{R}^\odot := \left\{ ((\mathbf{v}, \mu_1, \mu_2), (f_1, f_2), (\mathbf{f}_1, \mathbf{f}_2)) \mid \begin{array}{l} (f_1, f_2) = \mathcal{C}^{\equiv 2}(\mathbf{f}_1, \mathbf{f}_2) \\ \wedge \langle \mathbf{f}_1, \mathbf{v} \rangle = \mu_1 \wedge \langle \mathbf{f}_2, \mathbf{v} \rangle = \mu_2 \end{array} \right\}.$$

The verifier in the protocol has oracle access to the (purported) codewords, and aims to test whether they belong to the constrained code defined by $\mathcal{R}^\odot$.

To do so, the verifier requests a random linear combination of the messages, encodes the results, and checks uniformly at random that this encoding agrees with the inputs.

Construction 4.2. Let $\mathcal{C}: \mathbb{F}^k \rightarrow \mathbb{F}^n$ be a linear code.

- **Inputs.** The verifier receives as explicit input $\mathbf{x} = (\mathbf{v}, \mu_1, \mu_2)$ and as implicit input $\mathbf{y} = f_1, f_2: \mathcal{L} \rightarrow \mathbb{F}$. The honest prover receives additionally $\mathbf{w} = \mathbf{f}_1, \mathbf{f}_2 \in \mathbb{F}^k$ such that $\langle \mathbf{f}_1, \mathbf{v} \rangle = \mu_1$ and $\langle \mathbf{f}_2, \mathbf{v} \rangle = \mu_2$.
- **Interactive phase.**
 1. **Combination randomness.** The verifier samples and sends $\gamma \leftarrow \mathbb{F}$.
 2. The prover sends $\mathbf{g} \in \mathbb{F}^k$. In the honest case, $\mathbf{g} := \mathbf{f}_1 + \gamma \cdot \mathbf{f}_2$.
 3. **In domain randomness.** The verifier samples and sends $x_1, \dots, x_t \leftarrow \mathcal{L}$.
- **Decision phase.** The verifier computes $g := \mathcal{C}(\mathbf{g})$ and checks that:
 1. $\langle \mathbf{g}, \mathbf{v} \rangle = \mu_1 + \gamma \cdot \mu_2$.
 2. for $i \in [t]$ it holds that $g(x_i) = f_1(x_i) + \gamma \cdot f_2(x_i)$

We state (without proof) an upper bound on the round-by-round knowledge soundness error of Construction 4.2, a notion that we do not define (as unnecessary for this note) but that is sufficient for the protocol to be secure when compiled with the Fiat–Shamir transformation [FS86]. A complete proof can be found in [ABF26] (which also formally defines the mutual correlated agreement error ε_{mca} [ACFY25] which appears in the bound below).

Lemma 4.3. *Construction 4.2 has round-by-round knowledge soundness with errors*

$$\min_{\delta \in (0,1)} \left\{ \left(\varepsilon_{\text{mca}}(\mathcal{C}, \delta) + \frac{|\Lambda(\mathcal{C}, \delta)|}{|\mathbb{F}|}, (1 - \delta)^t \right) \right\}$$

4.2 List decoding attack

We show that the existence of a word with a large list of close codewords directly leads to an attack on the soundness of Construction 4.2. The attack exploits the fact that, if a codeword has a large list, the adversary, once the verifier has selected the random linear combination it desires, can choose any codeword in this list to maximize its success probability.

Lemma 4.4. *The soundness error of Construction 4.2 is at least*

$$\max_{\delta \in (0,1)} \frac{|\Lambda(\mathcal{C}, \delta)|}{|\mathbb{F}| + |\Lambda(\mathcal{C}, \delta)| - 1} \cdot (1 - \delta)^t.$$

Proof. We describe the attack next, fixing $\delta \in (0, 1)$.

- $\tilde{\mathbf{P}}(\delta)$:
 1. Let $f: \mathcal{L} \rightarrow \mathbb{F}$ be a word which attains the maximal list size and let $S := \{\mathbf{f} : \mathcal{C}(\mathbf{f}) \in \Lambda(\mathcal{C}, f, \delta)\}$ denote the messages underlying the codewords in the list.
 2. Let $\mathbf{v} \in \mathbb{F}^k$ be a vector maximizing the size of $S_{\mathbf{v}} := \{\langle \mathbf{f}, \mathbf{v} \rangle : \mathbf{f} \in S\}$ and let $\mu \in \mathbb{F} \setminus S_{\mathbf{v}}$.
 3. Initiate Construction 4.2 with as input $\mathbf{x} = (\mathbf{v}, 0, \mu)$, and implicit input $\mathbf{y} = (0^n, f)$.
 4. $\mathbf{V}$ samples $\gamma \leftarrow \mathbb{F}$.
 5. If there exists $\mathbf{g} \in S$ such that $\langle \mathbf{g}, \mathbf{v} \rangle = \gamma \cdot \mu$, reply with $\mathbf{g}$. Else return 0.
 6. $\mathbf{V}$ samples $x_1, \dots, x_t \leftarrow \mathcal{L}$, if for every $i \in [t]$ $\mathcal{C}(\mathbf{g})(x_i) = f(x_i)$ return 1, else return 0.

We analyze the probability that $\tilde{\mathbf{P}}(\delta) = 1$ (which is exactly when $\mathbf{V}$ accepts). First note that by construction $\Delta(\mathbf{y}, \mathcal{R}_{\mathcal{C}}[\mathbf{x}]) > \delta$. Note that in Item 5 the probability that the prover outputs 0 is exactly the probability that $\gamma \notin \{\frac{1}{\mu} : \mu \in S_{\mathbf{v}}\}$. Call this event E . Since by Claim 2.3 (applied to the identity zero-evader) $S_{\mathbf{v}}$ has size at least $\frac{|\mathbb{F}| \cdot |\Lambda(\mathcal{C}, \delta)|}{|\mathbb{F}| + |\Lambda(\mathcal{C}, \delta)| - 1}$, it holds that $\Pr[E] \geq |\Lambda(\mathcal{C}, \delta)| / (|\mathbb{F}| + |\Lambda(\mathcal{C}, \delta)| - 1)$. Conditioning on E , there exists $\mathbf{g} \in S$ such that $\langle \mathbf{g}, \mathbf{v} \rangle = \gamma \cdot \mu$. Since $\mathbf{g} \in S$, it must be that $\Delta(\mathcal{C}(\mathbf{g}), f) \leq \delta$ and so the probability that the $\tilde{\mathbf{P}}$ outputs 1 is at least $(1 - \delta)^t$. Therefore,

$$\begin{aligned} \Pr[\tilde{\mathbf{P}}(\delta) = 1] &= \Pr[\tilde{\mathbf{P}}(\delta) = 1 | E] \cdot \Pr[E] + \Pr[\tilde{\mathbf{P}}(\delta) = 1 | \neg E] \cdot \Pr[\neg E] \\ &\geq \frac{|\Lambda(\mathcal{C}, \delta)|}{|\mathbb{F}| + |\Lambda(\mathcal{C}, \delta)| - 1} \cdot (1 - \delta)^t \end{aligned}$$

□

As a corollary of Corollary 3.2 and Lemma 2.10 we obtain our main result.

Corollary 4.5. *The soundness error of Construction 4.2 is at least*

$$\max_{\delta \in (0,1)} \frac{|\mathbb{F}|^{n \cdot (\rho - 1 + H_{|\mathbb{F}|}(\delta))}}{(|\mathbb{F}| - 1) \sqrt{8 \cdot n \cdot \delta \cdot (1 - \delta)} + |\mathbb{F}|^{n \cdot (\rho - 1 + H_{|\mathbb{F}|}(\delta))}} \cdot (1 - \delta)^t.$$

If $\mathcal{C}$ is the extension code of a linear code $\mathcal{C}_{\mathbb{B}}: \mathbb{B}^k \rightarrow \mathbb{B}^n$ then the soundness error is at least

$$\max_{\delta \in (0,1)} \frac{|\mathbb{B}|^{n \cdot (\rho - 1 + H_{|\mathbb{B}|}(\delta))}}{(|\mathbb{B}| - 1) \sqrt{8 \cdot n \cdot \delta \cdot (1 - \delta)} + |\mathbb{B}|^{n \cdot (\rho - 1 + H_{|\mathbb{B}|}(\delta))}} \cdot (1 - \delta)^t.$$

Remark 4.6. It is natural to wonder whether the malicious prover $\tilde{\mathbf{P}}$ in Lemma 4.4 can be implemented in polynomial time. The barriers are represented by Items 1 and 2 and Item 5:

- Items 1 and 2 rely on the non-constructive proofs of Corollary 3.2 and Item 2. It seems reasonable to heuristically conjecture that sampling f and $\mathbf{v}$ uniformly at random will yield results approximately matching the lower bounds. A formal analysis of the second moment of the expected values to estimate the success probability of this approach is left as (near) future work.
- Item 5 requires finding a $\mathbf{g}$ such that $\mathcal{C}(\mathbf{g}) \in \Lambda(\mathcal{C}, f, \delta)$ and such that $\langle \mathbf{g}, \mathbf{v} \rangle = \gamma \cdot \mu$. This seems computationally unfeasible. In the range of parameters under which we mount the attacks, the size of the list is extremely large (exponential in the alphabet size), and a computationally efficient adversary cannot afford iterating through the list. Further, even if we were to allow the adversary to store $\Lambda(\mathcal{C}, f, \delta)$, computing the list-size seems to require some notion of list-decoding at (or over) capacity.

We remain uneasy at the possibility of a computationally efficient implementation of $\tilde{\mathbf{P}}$ or variations thereof.

Comparison with alternative toy protocol. [Sta21] introduces an alternative toy protocol, to which our attack does not apply. In the protocol therein the verifier:

1. has oracle access to $f: \mathcal{L} \rightarrow \mathbb{F}$ (which is claimed to be a Reed–Solomon codeword);
2. samples $\gamma \leftarrow \mathbb{F}$; and
3. tests whether the function $\frac{f(X) - \gamma}{X}$ is (close to) a Reed–Solomon codeword.

This toy protocol is purposefully chosen to not be perfectly complete, as the quotient is guaranteed to be a Reed–Solomon only when $\hat{f}(0) = \gamma$ which happens with probability at most $\frac{1}{|\mathbb{F}|}$. Soundness-wise, denoting by $\varepsilon_{\text{idt}}(\delta)$ the soundness error of the test used when the input function has distance δ from the Reed–Solomon code, it is easy to show that the protocol has round-by-round soundness errors $\leq \min_{\delta \in (0,1)} \left(\frac{|\Lambda(\mathcal{C}, \delta)|}{|\mathbb{F}|}, \varepsilon_{\text{idt}}(\delta) \right)$. Our attack does not apply to this toy protocol from [Sta21], as a large list-size $\Lambda(\mathcal{C}, f, \delta)$ does not guarantee that the set $\{\hat{g}(0) : \mathcal{C}(\hat{g}) \in \Lambda(\mathcal{C}, f, \delta)\}$ is of non-unit size. In particular, Claim 2.3 does not apply as the zero-evader error of the mapping $\hat{g} \mapsto \hat{g}(0)$ is $\varepsilon_{\text{zero}} = 1$. To the best of our understanding, it might very well be the case that all the codewords in lists of a sufficient size have the same constant terms. It seems plausible that variations of our attacks apply to modifications of the [Sta21] toy problem (for example, when the quotient is replaced by $\frac{f(X) - \alpha}{X - \beta}$), but we did not investigate this further.

4.3 Concrete estimate

We investigate the concrete security impact of Corollary 4.5 in concrete parameterization of the toy protocol (Construction 4.2).

A worked example. We instantiate Construction 4.2 in the setting where:

- $\mathbb{F}$ is a sextic extension of $\mathbb{B} = \mathbb{F}_q$ where $q = 2^{31} - 2^{24} + 1$;
- $k = 2^{20}$, $n = 2^{21}$ so that $\rho = 1/2$; and
- $\mathcal{L} \subseteq \mathbb{B}$ is a smooth domain with $|\mathcal{L}| = n$.
- $\mathcal{C} := \text{RS}[\mathbb{F}, \mathcal{L}, k]$ and $\mathcal{C}_{\mathbb{B}} := \text{RS}[\mathbb{B}, \mathcal{L}, k]$.
- $t = 128$.

This instantiation closely resembles concretely deployed instances of hash-based proof systems. We estimate the success probability of the prover $\tilde{\mathbf{P}}$ described in Lemma 4.4 in Construction 4.2 against this instantiation. By Corollary 4.5 we know that the success probability is at least

$$\begin{aligned} & \max_{\delta \in (0,1)} \frac{|\mathbb{B}|^{n \cdot (\rho - 1 + H_{|\mathbb{B}|}(\delta))} \cdot (1 - \delta)^t}{(|\mathbb{F}| - 1) \sqrt{8 \cdot n \cdot \delta \cdot (1 - \delta)} + |\mathbb{B}|^{n \cdot (\rho - 1 + H_{|\mathbb{B}|}(\delta))}} \\ &= \max_{\delta \in (0,1)} \frac{2^{31 \cdot 2^{21} \cdot (1/2 + H_{2^{31}}(\delta))} \cdot (1 - \delta)^{128}}{(2^{31 \cdot 6} - 1) \sqrt{8 \cdot 2^{21} \cdot \delta \cdot (1 - \delta)} + 2^{31 \cdot 2^{21} \cdot (1/2 + H_{2^{31}}(\delta))}} \\ &\geq 2^{-116.5} \end{aligned}$$

where the last value is obtained by a numerical search. Note that, as expected, the value at which the success probability is maximized $\approx 1 - \rho - \frac{1}{\log |\mathbb{B}|} = 0.4678 \dots$.

Other instantiations. In Table 1 we numerically estimate the success probability of Corollary 4.5 under various configurations of Construction 4.2. We consider different rates and size of the base field. As expected, smaller base fields $\mathbb{B}$ and larger rate ρ yield improved attacks. The scripts used to obtain these estimates are available at <https://gist.github.com/asanso/51100aea720b081dc5a119cbc7042627>.

$\log_2 \mathbb{B} $		16 ($e = 9$)	31 ($e = 5$)	64 ($e = 3$)
ρ				
$1/2$		(0.44, -106.58, 21.42)	(0.47, -116.55, 11.45)	(0.48, -122.36, 5.64)
$1/4$		(0.69, -109.63, 18.37)	(0.72, -118.52, 9.48)	(0.74, -123.48, 4.52)
$1/8$		(0.83, -111.75, 16.25)	(0.86, -120.19, 7.81)	(0.87, -124.88, 3.12)
$1/16$		(0.91, -111.38, 16.62)	(0.93, -119.84, 8.16)	(0.93, -124.05, 3.95)

Table 1: Estimate of success probability of Lemma 4.4 under different regimes. Each cell consists of $(\delta, \varepsilon, 128 - \varepsilon)$ where δ is the distance at which the success probability is maximized and ε is the success probability. t is always set to $\frac{128}{-\log \rho}$.

Acknowledgements

We thank Alistair Stewart for useful discussions, and Gal Arnon, Dan Boneh and Ziyi Guan for many helpful comments and suggestions.

References

- [ABF26] Gal Arnon, Dan Boneh, and Giacomo Fenzi. *Open Problems in List-Decoding and Correlated Agreement*. Cryptology ePrint Archive, Report 2026/680. 2026.
- [ACFY24] Gal Arnon, Alessandro Chiesa, Giacomo Fenzi, and Eylon Yogev. “STIR: Reed–Solomon Proximity Testing with Fewer Queries”. In: *Proceedings of the 44th Annual International Cryptology Conference*. CRYPTO ’24. 2024, pp. 380–413.
- [ACFY25] Gal Arnon, Alessandro Chiesa, Giacomo Fenzi, and Eylon Yogev. “WHIR: Reed–Solomon Proximity Testing with Super-Fast Verification”. In: *Proceedings of the 44th Annual International Conference on Theory and Application of Cryptographic Techniques*. EUROCRYPT ’25. 2025.
- [AHIV17] Scott Ames, Carmit Hazay, Yuval Ishai, and Muthuramakrishnan Venkitasubramaniam. “Ligero: Lightweight Sublinear Arguments Without a Trusted Setup”. In: *Proceedings of the 24th ACM Conference on Computer and Communications Security*. CCS ’17. 2017, pp. 2087–2104.
- [BBHR18] Eli Ben-Sasson, Iddo Bentov, Yinon Horesh, and Michael Riabzev. “Fast Reed–Solomon Interactive Oracle Proofs of Proximity”. In: *Proceedings of the 45th International Colloquium on Automata, Languages and Programming*. ICALP ’18. 2018, 14:1–14:17.
- [BCFRRZ25] Martijn Brehm, Binyi Chen, Ben Fisch, Nicolas Resch, Ron D. Rothblum, and Hadas Zeilberger. “Blaze: Fast SNARKs from Interleaved RAA Codes”. In: *Proceedings of the 44th Annual International Conference on Theory and Application of Cryptographic Techniques*. EUROCRYPT ’25. 2025.
- [BCFW25] Benedikt Bünz, Alessandro Chiesa, Giacomo Fenzi, and William Wang. *Linear time accumulation schemes*. Cryptology ePrint Archive, Report 2025/753. 2025.
- [BCIKS20] Eli Ben-Sasson, Dan Carmon, Yuval Ishai, Swastik Kopparty, and Shubhangi Saraf. “Proximity Gaps for Reed–Solomon Codes”. In: *Proceedings of the 61st Annual IEEE Symposium on Foundations of Computer Science*. FOCS ’20. 2020, pp. 900–909.
- [BFHVXZ20] Rishabh Bhaduria, Zhiyong Fang, Carmit Hazay, Muthuramakrishnan Venkitasubramaniam, Tiancheng Xie, and Yupeng Zhang. “Ligero++: A New Optimized Sublinear IOP”. In: *Proceedings of the 2020 ACM SIGSAC Conference on Computer and Communications Security*. CCS ’20. New York, NY, USA: Association for Computing Machinery, 2020, pp. 2025–2038.
- [BFRW25] Benedikt Bünz, Giacomo Fenzi, Ron D. Rothblum, and William Wang. *TensorSwitch: Nearly Optimal Polynomial Commitments from Tensor Codes*. IACR Cryptology ePrint Archive, Report 2025/2065. 2025.
- [BGKS20] Eli Ben-Sasson, Lior Goldberg, Swastik Kopparty, and Shubhangi Saraf. “DEEP-FRI: Sampling Outside the Box Improves Soundness”. In: *Proceedings of the 11th Innovations in Theoretical Computer Science Conference*. ITCS ’20. 2020, 5:1–5:32.
- [BMMS25a] Anubhav Baweja, Pratyush Mishra, Tushar Mopuri, and Matan Shtepel. *FICS and FACS: Fast IOPs and Accumulation via Code-Switching*. Cryptology ePrint Archive, Report 2025/737. 2025.

- [BMMS25b] Anubhav Baweja, Pratyush Mishra, Tushar Mopuri, and Matan Shtepel. *Query-Optimal IOPPs for Linear-Time Encodable Codes*. IACR Cryptology ePrint Archive, Report 2025/1588. 2025.
- [BMNW25a] Benedikt Bünz, Pratyush Mishra, Wilson Nguyen, and William Wang. “Accumulation Without Homomorphism”. In: *Proceedings of the 16th Innovations in Theoretical Computer Science Conference*. ITCS ’25. 2025, 23:1–23:25.
- [BMNW25b] Benedikt Bünz, Pratyush Mishra, Wilson Nguyen, and William Wang. “Arc: Accumulation for Reed–Solomon Codes”. In: *Proceedings of the 45th Annual International Cryptology Conference*. CRYPTO ’25. 2025.
- [BR93] Mihir Bellare and Phillip Rogaway. “Random Oracles are Practical: A Paradigm for Designing Efficient Protocols”. In: *Proceedings of the 1st ACM Conference on Computer and Communications Security*. CCS ’93. 1993, pp. 62–73.
- [CS25] Elizabeth Crites and Alistair Stewart. *On Reed–Solomon Proximity Gaps Conjectures*. Cryptology ePrint Archive, Paper 2025/2046. 2025. URL: <https://eprint.iacr.org/2025/2046>.
- [DG25] Benjamin E. Diamond and Angus Gruen. *On the Distribution of the Distances of Random Words*. Cryptology ePrint Archive, Paper 2025/2010. 2025. URL: <https://eprint.iacr.org/2025/2010>.
- [DP24] Benjamin E. Diamond and Jim Posen. “Proximity Testing with Logarithmic Randomness”. In: *IACR Communications in Cryptology* 1.1 (2024).
- [DP25] Benjamin Diamond and Jim Posen. “Succinct Arguments over Towers of Binary Fields”. In: *Proceedings of the 44th Annual International Conference on Theory and Application of Cryptographic Techniques*. EUROCRYPT ’25. 2025.
- [E25] Ethereum Foundation. *The Proximity Prize*. <https://proximityprize.org/>. 2025.
- [Eli57] Peter Elias. “List decoding for noisy channels”. In: (1957).
- [FS86] Amos Fiat and Adi Shamir. “How to prove yourself: practical solutions to identification and signature problems”. In: *Proceedings of the 6th Annual International Cryptology Conference*. CRYPTO ’86. 1986, pp. 186–194.
- [GGR11] Parikshit Gopalan, Venkatesan Guruswami, and Prasad Raghavendra. “List Decoding Tensor Products and Interleaved Codes”. In: *SIAM Journal on Computing* 40.5 (2011). Preliminary version appeared in STOC ’09., pp. 1432–1462.
- [GLSTW23] Alexander Golovnev, Jonathan Lee, Srinath T. V. Setty, Justin Thaler, and Riad S. Wahby. “Brakedown: Linear-time and field-agnostic SNARKs for R1CS”. In: *Proceedings of the 43rd Annual International Cryptology Conference*. CRYPTO ’23. 2023.
- [GMW25] Albert Garreta, Nicolas Mohnblatt, and Benedikt Wagner. *A Simplified Round-by-round Soundness Proof of FRI*. Cryptology ePrint Archive, Paper 2025/1993. 2025.
- [MS77] Florence Jessie MacWilliams and Neil James Alexander Sloane. *The theory of error-correcting codes*. Vol. 16. Elsevier, 1977.
- [MZ25] Dor Minzer and Kai Zhe Zheng. “Improved Round-by-round Soundness IOPs via Reed–Muller Codes”. In: *Proceedings of the 64th Annual IEEE Symposium on Foundations of Computer Science*. FOCS ’25. 2025.
- [Mic00] Silvio Micali. “Computationally Sound Proofs”. In: *SIAM Journal on Computing* 30.4 (2000). Preliminary version appeared in FOCS ’94., pp. 1253–1298.
- [NA25] Andrija Novakovic and Guillermo Angeris. *Ligerito: A Small and Concretely Fast Polynomial Commitment Scheme*. Cryptology ePrint Archive, Report 2025/1187. 2025.

- [RS60] I. S. Reed and G. Solomon. “Polynomial Codes Over Certain Finite Fields”. In: *Journal of the Society for Industrial and Applied Mathematics* 8.2 (1960), pp. 300–304.
- [RVW13] Guy N. Rothblum, Salil P. Vadhan, and Avi Wigderson. “Interactive proofs of proximity: delegating computation in sublinear time”. In: *Proceedings of the 45th ACM Symposium on the Theory of Computing*. STOC ’13. 2013, pp. 793–802.
- [Sta21] StarkWare. *ethSTARK Documentation*. Cryptology ePrint Archive, Paper 2021/582. <https://eprint.iacr.org/2021/582>. 2021. URL: <https://eprint.iacr.org/2021/582>.
- [WHV24] Ruihan Wang, Carmit Hazay, and Muthuramakrishnan Venkitasubramaniam. “Ligetron: Lightweight Scalable End-to-End Zero-Knowledge Proofs Post-Quantum ZK-SNARKs on a Browser”. In: *2024 IEEE Symposium on Security and Privacy (SP)*. 2024, pp. 1760–1776.
- [ZCF24] Hadas Zeilberger, Binyi Chen, and Ben Fisch. “BaseFold: Efficient Field-Agnostic Polynomial Commitment Schemes from Foldable Codes”. In: *Proceedings of the 44th Annual International Cryptology Conference*. Vol. 14929. CRYPTO ’24. 2024, pp. 138–169.